

Domain Heating								
code	service	1						
H-1a	Heat emission control	Service group: Heat control - demand side						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Central automatic control (e.g. central thermostat)	1	0	1	1	1	0	0
level 2	Individual room control (e.g. thermostatic valves, or electronic controller)	2	0	2	2	2	0	0
level 3	Individual room control with communication between controllers and to BACS	2	0	2	3	2	1	0
level 4	Individual room control with communication and occupancy detection	3	0	2	3	2	1	0
Information sources								
		EN 15232						
code	service	0						
H-1b	Emission control for TABS (heating mode)	Service group: Heat control - demand side						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Central automatic control	1	0	1	1	1	0	0
level 2	Advanced central automatic control	1	0	1	2	2	0	0
level 3	Advanced central automatic control with intermittent operation and/or room temperature feedback control	2	0	2	3	2	1	1
level 4								
Information sources								

	Standard?	EN 15232						
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code	service	1
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H-1c	Control of distribution fluid temperature (supply or return air flow or water flow) - Similar function can be applied to the control of direct electric heating networks	Service group: Heat control - demand side
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Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Outside temperature compensated control	1	0	1	1	0	0	0
level 2	Demand based control	2	0	1	1	0	0	0
level 3	0	-	-	-	-	-	-	-
level 4	0							

	Information sources							
	Standard?	EN 15232						

code	service	0
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H-1d	Control of distribution pumps in networks	Service group: Heat control - demand side
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Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	On off control	1	0	0	0	0	0	0
level 2	Multi-Stage control	2	0	0	0	0	0	0
level 3	Variable speed pump control (pump unit (internal) estimations)	2	0	0	0	0	0	0
level 4	Variable speed pump control (external demand signal)	2	0	0	0	0	0	0

	Information sources							
	Standard?	EN 15232						

code	service	✗0						
H-1f	Thermal Energy Storage (TES) for building heating (excluding TABS)	Service group: Heat control - demand side						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Continuous storage operation	0	0	0	0	0	0	0
level 1	Time-scheduled storage operation	1	0	0	0	0	0	0
level 2	Load prediction based storage operation	2	1	0	0	0	0	0
level 3	Heat storage capable of flexible control through grid signals (e.g. DSM)	2	2	0	0	0	0	0
level 4	()							
Information sources								
		Standard?	EN 15232					
code	service	✓1						
H-2a	Heat generator control (all except heat pumps)	Service group: Control heat production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Constant temperature control	0	0	0	0	0	0	0
level 1	Variable temperature control depending on outdoor temperature	1	0	1	0	0	0	0
level 2	Variable temperature control depending on the load (e.g. depending on supply water temperature set point)	2	0	2	0	0	0	0
level 3	()	-	-	-	-	-	-	-
level 4	()							
Information sources								
		Standard?	EN 15232					

code	service	1						
H-2b	Heat generator control (for heat pumps)	Service group: Control heat production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	On/Off-control of heat generator	0	0	0	0	0	0	0
level 1	Multi-stage control of heat generator capacity depending on the load or demand (e.g. on/off of several compressors)	1	1	1	0	0	0	0
level 2	Variable control of heat generator capacity depending on the load or demand (e.g. hot gas bypass, inverter frequency control)	2	1	2	0	0	0	0
level 3	Variable control of heat generator capacity depending on the load AND external signals from grid	2	3	2	0	0	0	0
level 4								
Information sources								
		Standard?						

code	service	0						
H-2d	Sequencing in case of different heat generators	Service group: Control heat production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Priorities only based on running time	0	0	0	0	0	0	0
level 1	Control according to fixed priority list: e.g. based on rated energy efficiency	1	0	0	0	0	0	0
level 2	Control according to dynamic priority list (based on current energy efficiency, carbon emissions and capacity of generators, e.g. solar, geothermal heat, cogeneration plant, fossil fuels)	2	1	0	0	0	0	0
level 3	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions and capacity of generators)	3	2	0	0	0	0	0
level 4	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions, capacity of generators AND external signals from grid)	3	3	0	0	0	0	0
Information sources								
		Standard?	EN 15232					

code	service	✓1
H-3	Report information regarding heating system performance	Service group: Information to occupants and facility management

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Central or remote reporting of current performance KPIs (e.g. temperatures, submetering energy usage)	1	0	0	0	0	1	1
level 2	Central or remote reporting of current performance KPIs and historical data	1	0	0	0	0	1	2
level 3	Central or remote reporting of performance evaluation including forecasting and/or benchmarking	1	0	0	0	0	1	3
level 4	Central or remote reporting of performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	1	0	0	1	0	3	3
Information sources								
		Standard?						

code	service	✗0
H-4	Flexibility and grid interaction	Service group: Flexibility and grid interaction

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Scheduled operation of heating system	1	0	1	1	0	0	0
level 2	Self-learning optimal control of heating system	2	1	2	2	0	0	0
level 3	Heating system capable of flexible control through grid signals (e.g. DSM)	2	3	2	3	0	0	0
level 4	Optimized control of heating system based on local predictions and grid signals (e.g. through model predictive control)	2	3	3	3	1	0	0